



(Don't Write
anything in this Area)

4.b) Explain the significance of Screening and Counselling for prevention of Genetic Disorders. (15 Marks)

Genetic Screening is the process to screen out the genes of the Human, who either has vulnerability of any autosomal or allosomal disease or can pass on these genes to next generation.

Genetic Counselling is the psychological anthropological way to reach out to the genetically screened person. to deliver the message of their health and disease to further control the spread of it.

Steps involved in the process :-

① Counsellor will collect the large amount of genetic data

② Understand and evaluate the individual genetic history.

↳ it can be done via the pedigree analysis and application of medical history.

③ Analyzing the pattern of transmission of the disease.

↳ whether it is dominant or recessive.

↳ whether it is more recurrent and in which generation.

④ ~~# help~~ Then the information needs to be communicated to the individuals/couple.

However, ethical consideration like confidentiality, consent and privacy has to be maintained to respect the human dignity.

Examples showing it used in
preventing the Genetic disorder

① CSIR initiative to screen the
CCR5 Δ32 → HIV resistant gene.

Found in North Indians it is
(0.7 - 5%) and South = 0.6% to 1.5%
↓
More HIV.

② National Mission to eliminate
sickle cell Anaemia among Tribals
by 2047.

③ JSCREEN & NAF launched screening
for Ashkenazi Jews suffering from
Gaucher disease.

④ Gujarat government prevented the
birth of "182" Thalassemic kids

⑤ In Saharā tribe → HLA DRB1*15 →
TB vulnerable; HLA DRB1*16 → TB
susceptible.

Thus, we can see how genetic
screening is working with counselling.